

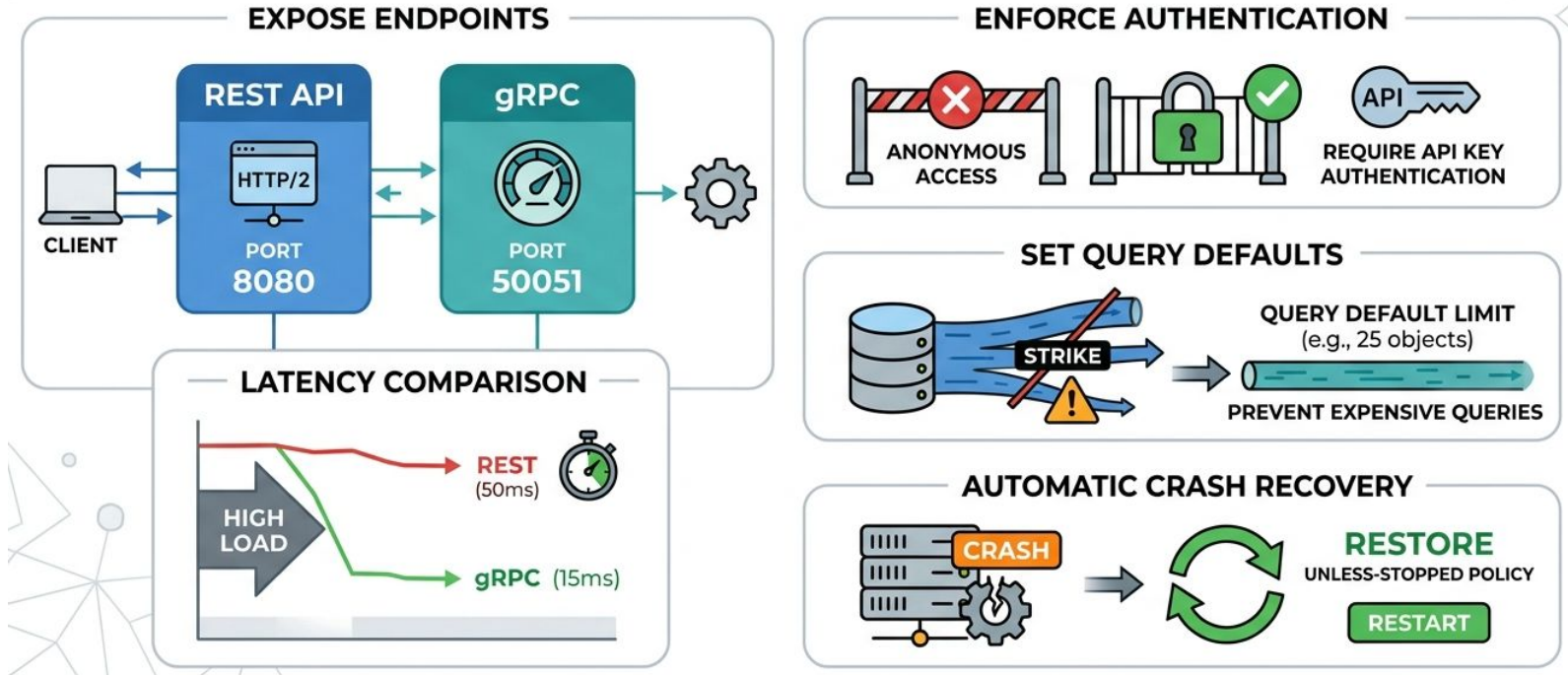


U.S. General Services
Administration

Chapter 6.2B: Production Vector Database Deployment

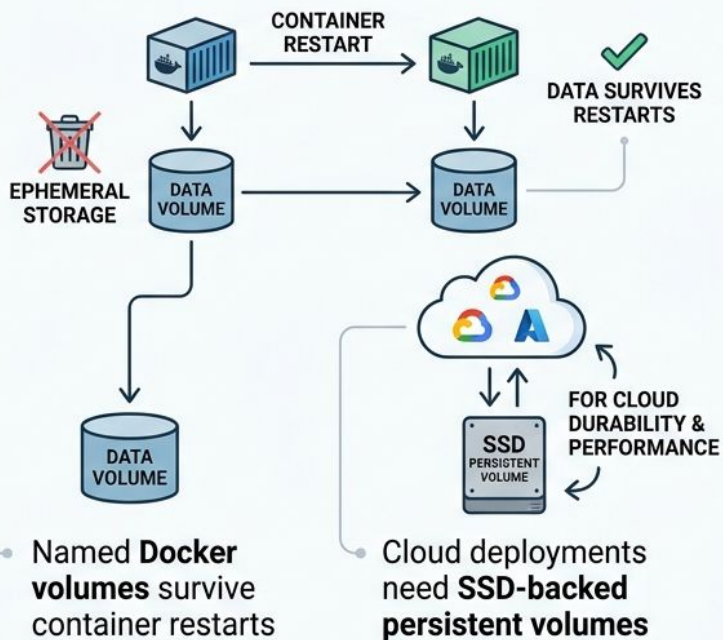
US General Service Administration
AI Community of Practice - March 2026

Docker Compose Production Configuration



Persistence and Resource Management

DURABILITY & PERSISTENCE

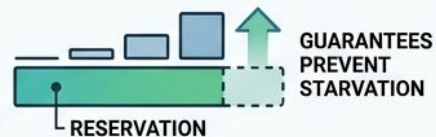


RESOURCE MANAGEMENT

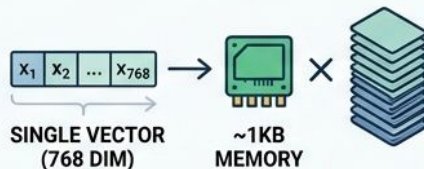
Set resource limits: prevent runaway consumption



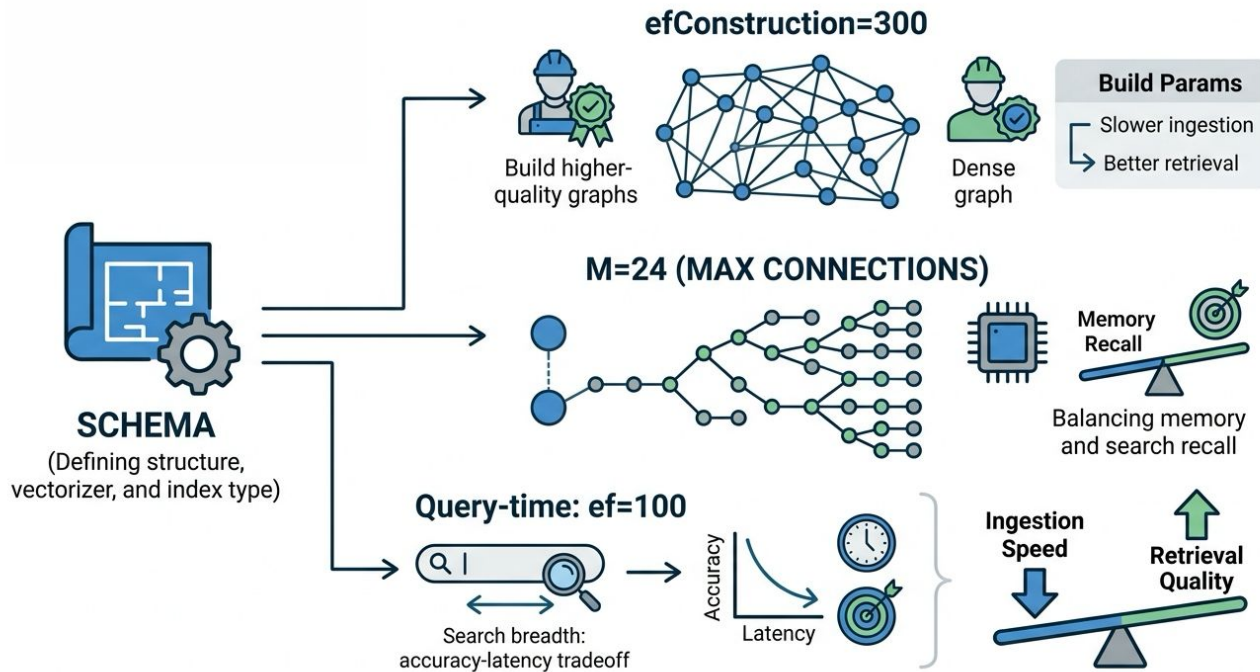
Set resource reservations: guarantees prevent starvation



Rule of thumb:
~1KB memory per 768-dimensional vector

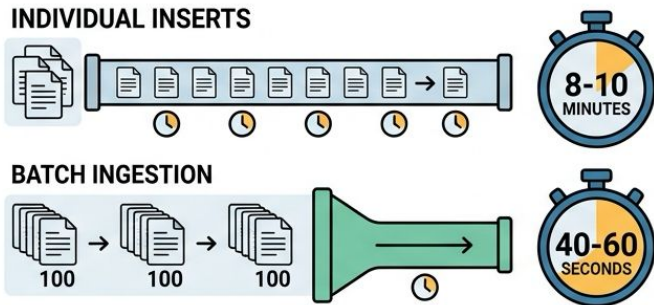


Schema Design and HNSW Tuning

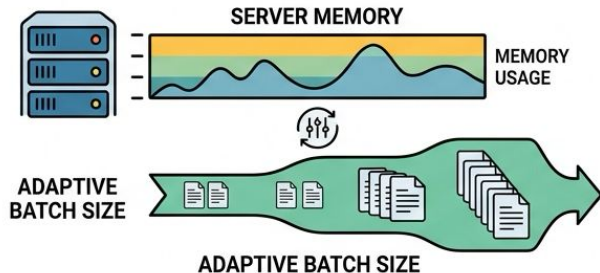


Batch Ingestion at Production Scale

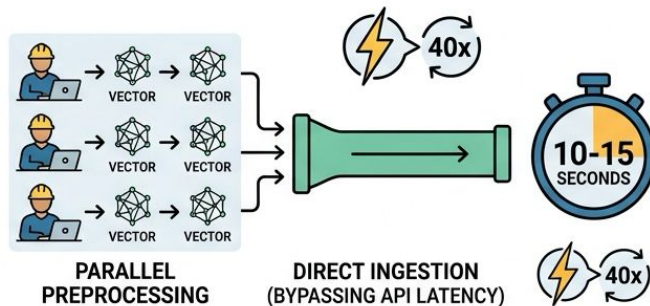
INGESTION APPROACHES



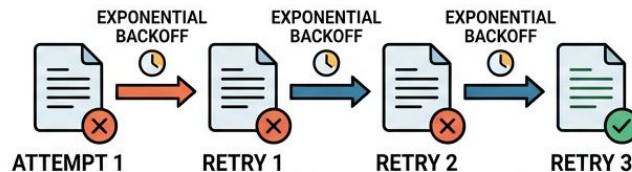
DYNAMIC BATCHING



OPTIMIZING WITH PRE-COMPUTED VECTORS



RETRY LOGIC & BACKOFF

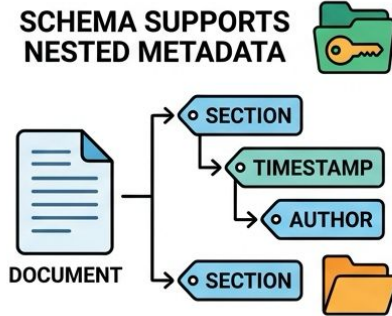


Hybrid Search -- Best of Both Worlds

- Pure vector search misses exact terms; pure keyword misses semantics
 - Hybrid search fuses BM25 and vector scores for 15-25% accuracy gain
 - The alpha parameter controls vector-vs-keyword balance
 - $\alpha=0.7$ suits conceptual queries; $\alpha=0.3$ suits technical lookups
 - Adds only 20-40% latency over pure vector search
- 

Metadata Filtering and Source Attribution

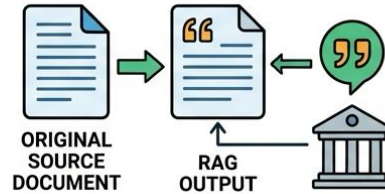
SCHEMA SUPPORTS NESTED METADATA



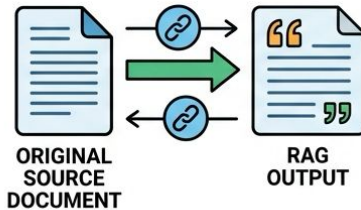
FILTER QUERIES BY METADATA FIELDS FOR PRECISE RETRIEVAL



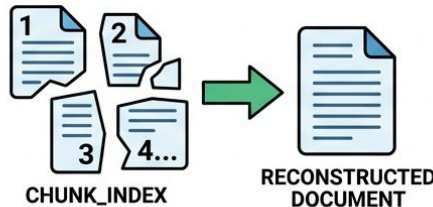
SOURCE TRACKING ENABLES CITATION AND PROVENANCE IN RAG OUTPUT



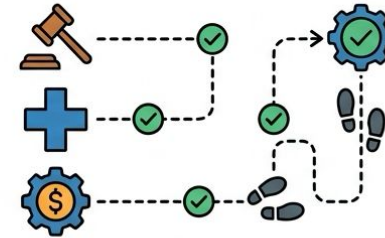
SOURCE TRACKING ENABLES CITATION AND PROVENANCE IN RAG OUTPUT



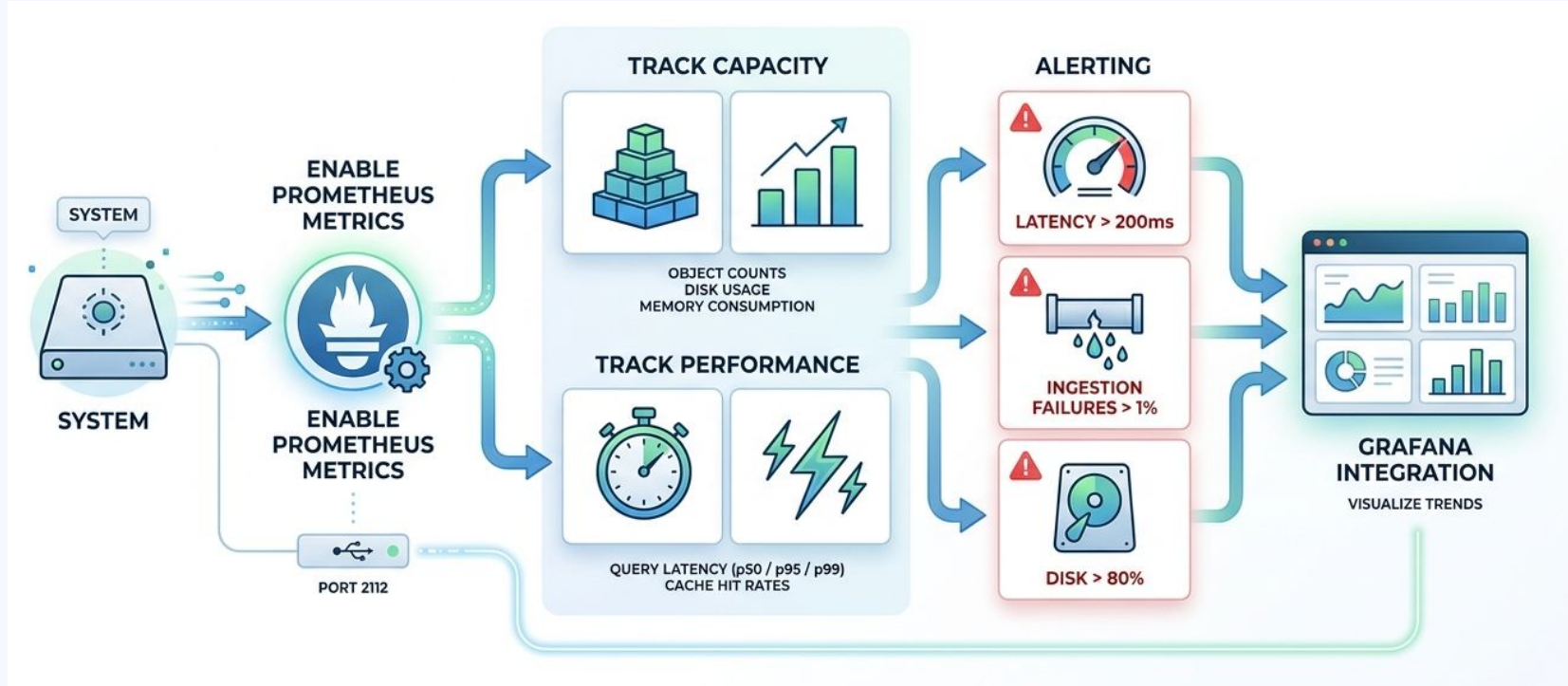
'CHUNK_INDEX' PRESERVES DOCUMENT ORDERING FOR RECONSTRUCTION



COMPLIANCE DOMAINS REQUIRE ATTRIBUTION TRAILS

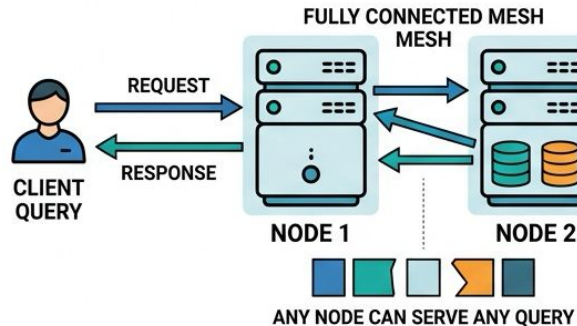
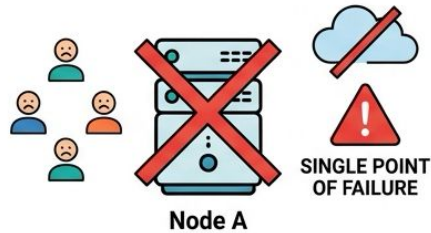


Monitoring and Alerting in Production

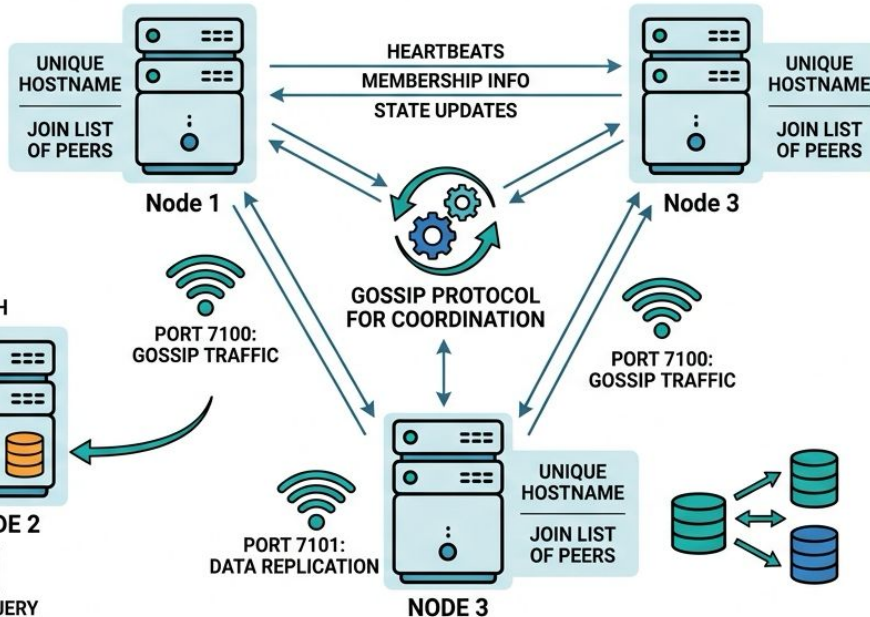


High Availability with Clustered Deployment

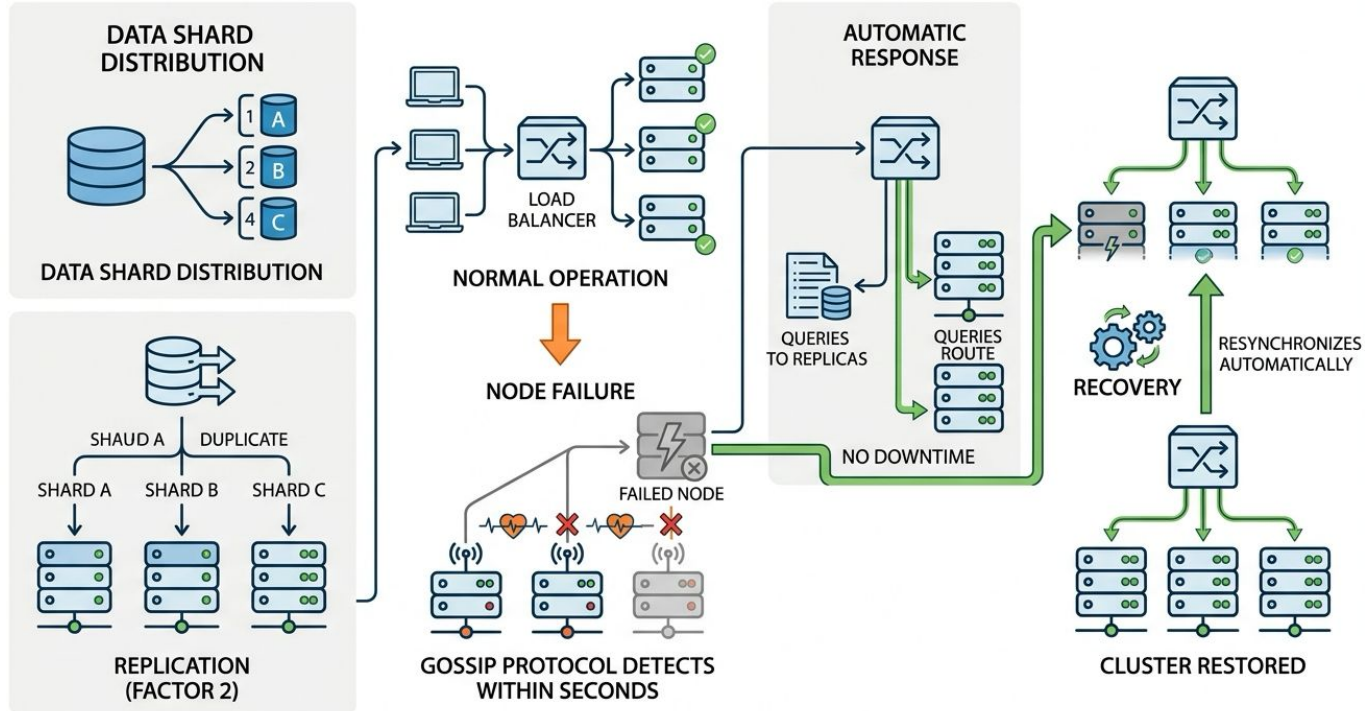
Single-Node Deployment



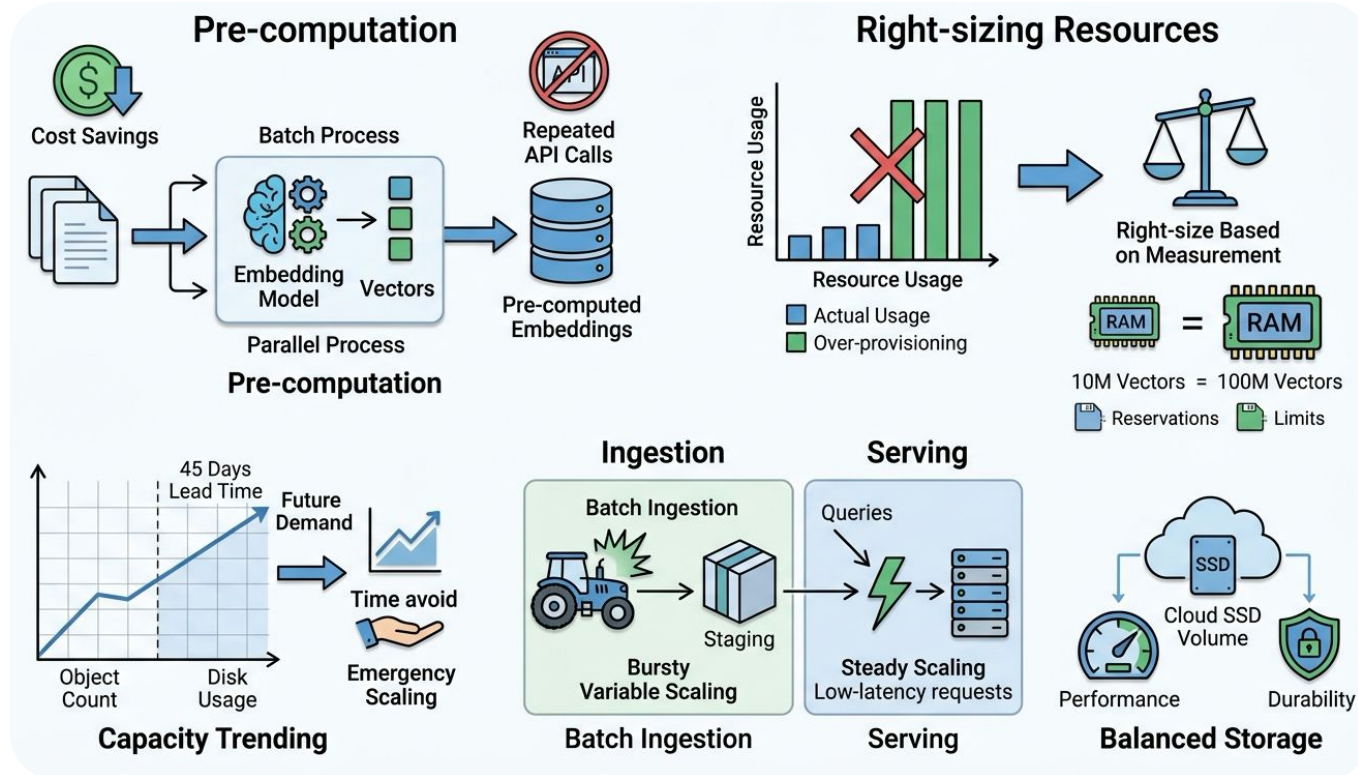
Three-Node Cluster Deployment



Replication and Fault Tolerance



Cost Optimization and Scaling Decisions



Key Takeaways

- Production vector DBs layer security, persistence, performance, observability
 - Batch ingestion and hybrid search are non-negotiable at scale
 - Clustered deployments eliminate single points of failure
 - Monitoring transforms operations from reactive to proactive
 - Next: Chapter 6.3A builds the ETL pipelines that feed these databases
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